

Genetic-precision indications identified by hereditary, mutation, congenital, familial, genetic, or deficiency terminology predict lower repurposing success.

AUDIT NUMBERS — FROM THE REGISTRY AT SAVE TIME

FRAME	TEST	OR	95% CI	N	DISCOVERY RAW P	FDR Q	CONFIRM RAW P
narrow	fisher_exact	0.391	[0.0868, 1.18]	2913	1.98e-1	4.09e-1	—
broad	fisher_exact	3.04	[1.32, 6.24]	4203	3.93e-3	1.31e-2	2.38e-2

CONFOUND	UNADJ. OR	ADJ. OR	ADJ. 95% CI	ADJ. P	SURVIVES?
rare-disease indication prevalence	3.04	3.14	[1.4057, 7.0091]	5.26e-3	survives
prior repurposing track record depletion	3.04	4.90	[2.186, 10.9713]	1.13e-4	survives
established-product maturity	3.04	2.93	[1.316, 6.5381]	8.50e-3	survives

Repurposing-Success Association Report: Genetic-Precision Indication Terminology

Hypothesis

The hypothesis, proposed by Sol, states verbatim: "*Genetic-precision indications identified by hereditary, mutation, congenital, familial, genetic, or deficiency terminology predict lower repurposing success.*" The stated source domain is "**kinetic proofreading under near-cognate T-cell receptor recognition.**"

Prior-literature tag (web-search, informational): NOVEL

Bisociative provenance

Opus proposed:

- impedance matching in transmission-line power transfer
- loss aversion / endowment effect in behavioral decision theory
- quorum sensing thresholds in bacterial gene regulation
- stigmergy / pheromone-trail reinforcement in ant foraging
- vestibular sensory reweighting under conflicting cues (multisensory integration)

Sol proposed:

- cycle-life dominance in rechargeable batteries under repeated duty cycles
- impedance matching by transformer networks in radio-frequency circuits
- kinetic proofreading under near-cognate T-cell receptor recognition
- mode matching at refractive-index discontinuities in optical waveguides

Lead reviewer consolidation (from lead_decisions): The lead_decisions array records numerous per-hypothesis dispositions but does not use the labels READY, NEEDS_ENRICHMENT, or DISCARDED. Rather than fabricate those categories, I report the mechanical dispositions actually present. Many entries were computed effect summaries (e.g., the two "kinetic proofreading" entries for this hypothesis, reported at OR=0.391 CI[0.0868,1.18] n=2913 and OR=3.04 CI[1.32,6.24] n=4203, with the mechanistic rationale that genotype-defined disorders impose narrow causal requirements rejecting near-cognate mechanisms). Others were marked "**not tested**" for reasons such as *perfect separation within a moderator level* or *base is constant within a moderator level*; several were "**SKIPPED (duplicate)**" because an identical feature_spec had already been tested in another run; two were tagged "**SALVAGEABLE**" with notes about degenerate nested-threshold interactions. The explicit READY/NEEDS_ENRICHMENT/DISCARDED tri-state requested is **not present** in FACTS, so I cannot map each hypothesis to it.

How this was actually tested

The literal computable proxy (`how_tested`) was:

indication name (lowercased) contains any of: [mutation, mutant, hereditary, familial, congenital, genetic, deficiency]

This mechanical keyword-matching rule on the indication-name column — not the broader biological claim about genotype-imposed pharmacological rejection — is what the statistics below actually measured.

Statistical evidence underneath the p-value

FRAMING	TEST	OR	95% CI	N	DISCOVERY RAW P	FDR Q
narrow	fisher_exact	0.391	[0.0868, 1.18]	2913	0.19828176327093952	0.40886880477637033
broad	fisher_exact	3.04	[1.32, 6.24]	4203	0.003927811726406188	0.013050471219994753

Narrow framing: The odds ratio of 0.391 (below 1) points toward *lower* odds of repurposing success for keyword-tagged indications — directionally consistent with the hypothesis — but the confidence interval crosses 1 and the effect is roughly a ~60% reduction in odds; this framing was not statistically significant.

Broad framing: The odds ratio of 3.04 (above 1) indicates roughly *tripled* odds of repurposing success for keyword-tagged indications — a direction **opposite** to the hypothesized "lower success" — with a confidence interval entirely above 1.

Discovery vs. confirmation

FRAMING	DISCOVERY RAW P	FDR Q	DISCOVERY PASS	CONFIRMATION RAW P	CONFIRMATION PASS
narrow	0.19828176327093952	0.40886880477637033	false	null	false
broad	0.003927811726406188	0.013050471219994753	true	0.023763391402535312	true

The **broad** framing passed both discovery (FDR q = 0.0131) and independent holdout confirmation (raw p = 0.0238). Passing confirmation means the effect reproduced on data never used during discovery. The **narrow** framing did not pass discovery and has a null confirmation p-value (confirmation not applicable/absent).

Confound checks

`confound_check.status` is **completed**, with the caveat noted in FACTS: "*candidate explanations checked — not the confirmed mechanism.*" All three confounds below were computable, and each adjustment is applied to the primary effect (unadjusted OR=3.0443, p=0.003928, n=4203).

1. Rare-disease indication prevalence — Rationale: genetic/hereditary/deficiency terminology overwhelmingly tags rare orphan diseases with tiny patient populations, which independently depress repurposing success; the keyword feature may simply proxy rare-disease status.

Computable: **yes**
Unadjusted: OR = 3.0443, p = 0.003928
Adjusted: OR = 3.1389, 95% CI [1.4057, 7.0091], p = 0.005259
Survives adjustment: true

2. Prior repurposing track record depletion — Rationale: genetically-defined indications may attract fewer historical repurposing attempts, and low prior-attempt volume both correlates with the keyword class and independently predicts lower success; the effect could be driven by prior-attempt volume.

Computable: **yes**
Unadjusted: OR = 3.0443, p = 0.003928
Adjusted: OR = 4.8973, 95% CI [2.186, 10.9713], p = 0.000113
Survives adjustment: true

3. Established-product maturity — Rationale: drugs for genetic/deficiency indications may less often be established marketed products, and established products carry safety/dosing/manufacturing maturity that raises success regardless of genotype specificity.

Computable: **yes**

Unadjusted: OR = 3.0443, p = 0.003928

Adjusted: OR = 2.9333, 95% CI [1.316, 6.5381], p = 0.008501

Survives adjustment: true

Verdict: The primary effect survived adjustment for **every** computable confound tested — rare-disease prevalence, prior repurposing track record, and established-product maturity. In each case the adjusted odds ratio remained above 1 with a confidence interval excluding 1 and a significant p-value. No computable confound caused the effect to fail (not survive) adjustment.

Bottom line

The finding is a statistical association in a curated dataset: indications whose names contain the specified genetic-precision keywords are associated, under the **broad** framing, with roughly tripled odds of repurposing success (OR=3.04, confirmed on holdout at p=0.0238), robust to adjustment for three tested confounds. This is **not** evidence of causation, biological mechanism, or clinical relevance, and it directionally **contradicts** the original hypothesis of *lower* success (the significant/confirmed effect points to higher, not lower, odds). The strongest remaining caveat is that the effect is defined entirely by a lexical keyword proxy on indication names rather than any verified genetic/pharmacological property, and only candidate confounds present in FACTS were tested — unmeasured confounders remain possible.

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